Violent actors and their interactions. Forecasting conflict dynamics

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The golden triangle?

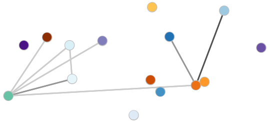
- Promise 1: Disaggregation instead of aggregation.
- Promise 2: Non-linearity vs. linearity; Non-parametric vs. parametric
- Promise 3: (Strategic) Interdependence instead of independence

Predictive turn in conflict studies

- Gurr and Lichbach, 1986
- O'brien, 2002
- Schrodt, 2006
- Goldstone et al., 2010
- Ward, 2010
- Weidmann and Ward, 2010
- Schneider, Gleditsch and Carey, 2011
- De Mesquita, 2011
- Ward et al., 2013
- Gleditsch and Ward, 2013
- Hegre et al., 2013
- Bell et al., 2013
- Brandt, Freeman and Schrodt, 2014
- Muchlinski et al., 2016

Interdependence turn

Monadic analysis → Dyadic analysis → Network analysis

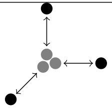
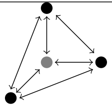


Collier and Hoeffler (2000)
Fearon and Laitin (2003)
Bates et al. (2005)
Cunningham (2006)
Gleditsch et al. (2002)
Hegre et al. (2001)

Cunningham et al. (2009)
Wucherpfennig et al. (2012)
Bapat and Bond (2012)
Walter (2009)
Bakke et al. (2012)
Steinwand (2011)
Cunningham (2013)

Ward et al. (2013)
Metternich et al. (2013)
Starr and Most (1983)
Maoz (2010)
Larson and Lewis (2017)
Chyzh (2016)
Franzese and Hays (2008)

Table: Ideal types of analyzing independence in conflicts.

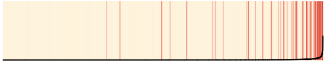
Type	Representation	Rebels' Utility	$y = \ldots + \epsilon$
Aggregate		$U_i(X)$	$X\beta$
Dyadic		$U_i(X, Z)$	$X\beta + Z\zeta$
K-dyads		$U_i(X, Z, K)$	$X\beta + Z\zeta + K\kappa$
K-adic		$U_i(X, Z, Y_j, W)$	$X\beta + Z\zeta + K\kappa + \rho Wy$

Example 1

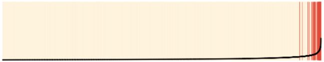
Metternich, Minhas, and Ward, 2017

FIGURE 1. *Conflict Contagion, 2001-2005.*

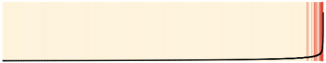
(A) Dyadic Logit Model (01-05)



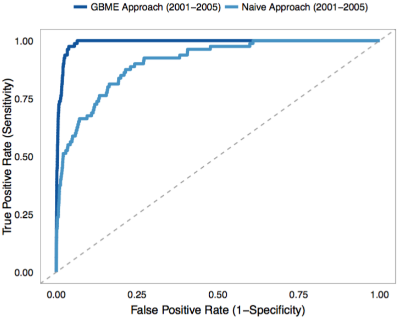
(B) Dyadic Logit Model (06-10)

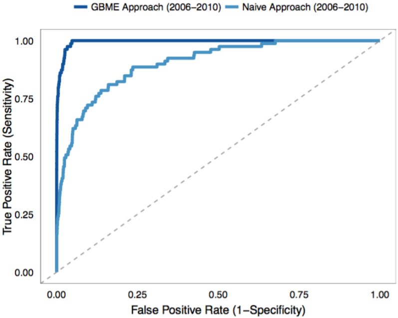


(C) GBME Model (01-05)



(D) GBME Model (06-10)





Example 2

Metternich and Wucherpfennig, working paper

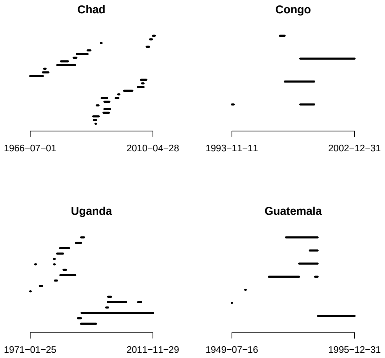
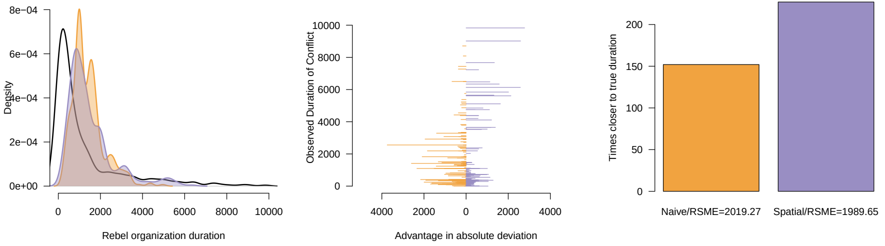


Figure: Rebel organization activity during selected UCDP armed conflicts. Each row represents a separate rebel organization as defined by a unique UCDP *SideB* for each rebel organization.

Model fit and insample prediction



(a) Predicted densities (b) Relative advantage (c) Case-wise comparison

Figure: Model fit and insample prediction assessment comparing Model 1 (k-dyadic Weibull controlling for the number of rebel organizations) and Model 3 (spatial k-adic Weibull including the number of rebel organizations). In all panels orange refers to Model 1, while purple pertains to Model 3.

Example 3

Metternich et al., conference paper

RebelCast:

- Makes predictions for rebel organizations (disaggregation)
- Supported by machine learning ensembles
- With a focus on interdependent actors

Conclusion

Suggestions for GCRI

- Disaggregate (but needs to be operationally relevant)
- Allow for non-linear, non-parametric processes
- Allow for interdependencies between actors (governments, states, rebel organizations etc)